

Experiment 3

Student Name: Yash Sharma

Branch: B.E. CSE

Semester: 5th

Subject Name: Full Stack Development - II

UID: 24BCS11074

Section/Group: 610-B

Date of Performance: July 2026

Subject Code: 24CSP-304

Easy

1. **Aim:** Set up React Router with three pages — Home, About, and Contact — navigated via a Navbar component using <Link>.

2.Objective:

- To understand client-side routing in React.
- To learn the installation and setup of React Router.
- To create multiple pages in a React application.
- To implement navigation using the <Link> component.
- To build a simple Single Page Application (SPA).

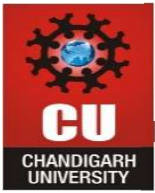
3.Code:

- **Navbar.jsx**

```
import { Link } from "react-router-dom";

function Navbar()
{
  return (
    <nav style={{ padding: "10px", background: "#ddd" }}>
      <Link to="/" style={{ marginRight: "10px" }}>Home</Link>
      <Link to="/about" style={{ marginRight: "10px" }}>About</Link>
      <Link to="/contact">Contact</Link>
      { /* <Link to = "/contact"> Contactssss </Link> */ }
    </nav>
  );
}

export default Navbar;
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

- **App.jsx**

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Navbar from "../components/Navbar";
import Home from "../pages/Home";
import About from "../pages/About";
import Contact from "../pages/Contact";
import "../App.css";
```

```
function App() {
  return (
    <BrowserRouter>
      <Navbar />
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
        <Route path="/contact" element={<Contact />} />
      </Routes>
    </BrowserRouter>
  );
}
```

```
export default App;
```

- **About.jsx**

```
function About() {
  return <h2>Welcome to About Page</h2>;
}
```

```
export default About;
```

- **Home.jsx**

```
function Home() {
  return <h2>Welcome to Home Page</h2>;
}
```

```
export default Home;
```

- **Contact.jsx**

```
function Contact() {
  return <h2>Welcome to Contact Page</h2>;
}
```

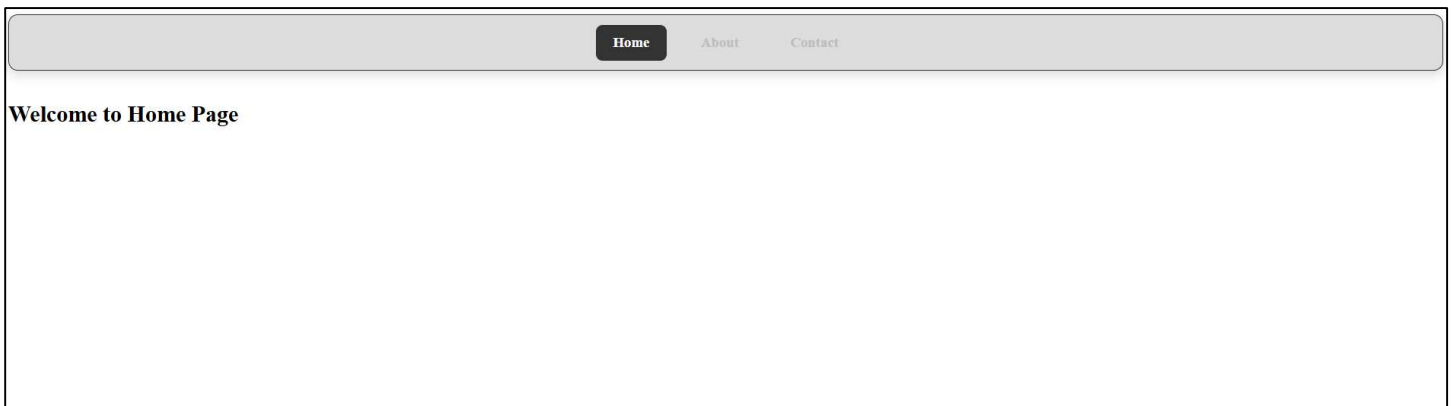
```
export default Contact;
```

- **Main.jsx**

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

4. Output:



5. Learning Outcome:

- Understanding of client-side routing concepts in React.
- Ability to install and configure React Router DOM.
- Skill in creating multiple page components and linking them with `<Link>`.
- Confidence in building a basic Single Page Application (SPA).
- Foundation for multi-page navigation without full page reloads.

Medium

1. **Aim:** Build a student directory SPA with routes for /students (list), /students/:id (detail page), and /add (form). Pass data via route params and display individual student profiles.

2.Objective:

- To understand dynamic routing in React Router.
- To implement route parameters using useParams.
- To create navigation between pages.
- To display dynamic student profile details.
- To build a basic student management SPA.

3.Code:

- **App.jsx**

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Navbar from "../components/Navbar";
import Students from "../pages/Students";
import StudentDetail from "../pages/StudentDetail";
import AddStudent from "../pages/AddStudent";

function App() {
  return (
    <BrowserRouter>
      <Navbar />
      <Routes>
        <Route path="/students" element={<Students />} />
        <Route path="/students/:id" element={<StudentDetail />} />
        <Route path="/add" element={<AddStudent />} />
      </Routes>
    </BrowserRouter>
  );
}

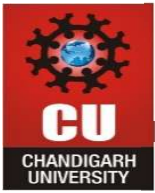
export default App;
```

- **StudentDetail.jsx**

```
import { useParams } from "react-router-dom";
import students from "../data/students";

function StudentDetail() {
  const { id } = useParams();
  const student = students.find((s) => s.id === parseInt(id));

  if (!student) {
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

```
    return <h2>Student Not Found</h2>;
  }

  return (
    <div>
      <h2>Student Profile</h2>
      <p><strong>ID:</strong> {student.id}</p>
      <p><strong>Name:</strong> {student.name}</p>
      <p><strong>Course:</strong> {student.course}</p>
      <p><strong>Age:</strong> {student.age}</p>
    </div>
  );
}

export default StudentDetail;
```

- **AddStudent.jsx**

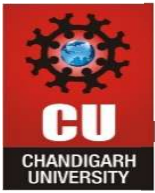
```
function AddStudent() {
  return (
    <div>
      <h2>Add Student Page</h2>
      <form>
        <input type="text" placeholder="Enter Name" />
        <br /><br />
        <input type="text" placeholder="Enter Course" />
        <br /><br />
        <input type="number" placeholder="Enter Age" />
        <br /><br />
        <button>Add Student</button>
      </form>
    </div>
  );
}

export default AddStudent;
```

- **main.jsx**

```
import React from "react";

import ReactDOM from "react-dom/client";
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

```
import App from "./App";
```

```
ReactDOM.createRoot(document.getElementById("root")).render(  
  

```

```
<React.StrictMode>
```

```
<App />
```

```
</React.StrictMode>
```

```
);
```

- **Students.jsx**

```
import { Link } from "react-router-dom";
```

```
import students from "../data/students";
```

```
function Students() {
```

```
  return (  
    <div>
```

```
      <h1>Student Directory</h1>
```

```
      {students.map((student) => (  
        <div key={student.id}>
```

```
          <Link to={`/${students}/${student.id}`}>
```

```
            {student.name}
```

```
          </Link>
```

```
        </div>
```

```
      </div>
```

```
    </div>
```

```
  ));
```

```
</div>
```

```
);
```

```
}
```

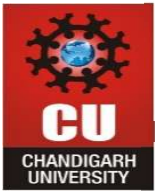
```
export default Students;
```

- **students.js**

```
const students = [  
  { id: 1, name: "Yash", course: "CSE", age: 20 },
```

```
  { id: 2, name: "Ranbir", course: "CSE", age: 21 },
```

```
  { id: 3, name: "Gaurav", course: "CSE", age: 21 }]
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

```
];
```

```
export default students;
```

4. Output:

[Students](#) | [Add Student](#)

Student Directory

[Yash](#)
[Ranbir](#)
[Gaurav](#)

5. Learning Outcome:

- Understanding of dynamic routing and route parameters in React Router.
- Ability to use useParams hook to extract URL parameters.
- Skill in mapping data to navigable links and rendering dynamic detail views.
- Confidence in structuring a multi-route SPA with list → detail navigation.
- Foundation for building data-driven student / employee directory applications.

Hard

1. **Aim:** Create a multi-page admin SPA with protected routes (login required), a dashboard with student statistics, nested routing for sub-sections, and a 404 Not Found page.

2.Objective:

- To understand advanced routing concepts in React Router.
- To implement protected routes using authentication logic.
- To create nested routing structures.
- To develop a dashboard with student statistics.
- To handle invalid routes using a 404 page.
- To improve application scalability and navigation structure.

3.Code:

- **Sidebar.jsx**

```
import { Link } from "react-router-dom";

function Sidebar({ handleLogout }) {
  return (
    <div
      style={{
        width: "200px",
        backgroundColor: "#f4f4f4",
        padding: "20px",
        height: "100vh",
        boxSizing: "border-box"
      }}
    >
      <h2 style={{ marginBottom: "20px" }}>Admin Panel</h2>
      <nav style={{ display: "flex", flexDirection: "column", gap: "10px" }}>
        <Link to="/dashboard/students" style={{ color: "blue" }}>Students</Link>
        <Link to="/dashboard/reports" style={{ color: "blue" }}>Reports</Link>
        <Link to="/dashboard/users" style={{ color: "blue" }}>Users</Link>
      </nav>

      <button
        onClick={handleLogout}
        style={{
          marginTop: "30px",
          padding: "8px 16px",
          backgroundColor: "#f44336",
          color: "white",
          border: "none",
          borderRadius: "5px",
          cursor: "pointer"
        }}
      >Logout
    </div>
  );
}
```



```

    }}
  >
    Logout
  </button>
</div>
);
}

export default Sidebar;

```

- **Dashboard.jsx**

```

import { Outlet, useNavigate } from "react-router-dom";
import Sidebar from "../components/Sidebar";
import { useEffect, useState } from "react";

function Dashboard({ setIsAuthenticated }) {
  const navigate = useNavigate();
  const [users, setUsers] = useState([]);

  function handleLogout() {
    setIsAuthenticated(false);
    navigate("/");
  }

  useEffect(() => {
    async function fetchUsers() {
      const res = await fetch("https://jsonplaceholder.typicode.com/users");
      const data = await res.json();
      setUsers(data);
    }
    fetchUsers();
  }, []);

  return (
    <div style={{ display: "flex" }}>
      { /* Sidebar */ }
      <Sidebar handleLogout={handleLogout} />

      { /* Main Content */ }
      <div style={{ flex: 1, padding: "20px" }}>
        <h1 style={{ marginBottom: "20px" }}>Admin Dashboard</h1>

        <h3 style={{ marginBottom: "10px" }}>Student Statistics</h3>
        <ul style={{ marginBottom: "20px", lineHeight: "1.8" }}>
          <li>Total Students: {users.length}</li>
          <li>Courses: 12</li>
          <li>Active Users: {Math.floor(users.length * 0.7)}</li>
        </ul>

        <div style={{ borderTop: "1px solid #ccc", paddingTop: "20px" }}>

```

```

        <Outlet />
      </div>
    </div>
  </div>
);
}

export default Dashboard;

```

- **Users.jsx**

```

import { useEffect, useState } from "react";

function Users() {
  const [users, setUsers] = useState([]);

  useEffect(() => {
    async function fetchUsers() {
      try {
        const res = await fetch("https://jsonplaceholder.typicode.com/users");
        const data = await res.json();
        setUsers(data);
      } catch (error) {
        console.error("Error fetching users:", error);
      }
    }
    fetchUsers();
  }, []);

  return (
    <div style={{ padding: "20px" }}>
      <h2 style={{ marginBottom: "15px" }}>Fetched Users</h2>
      <table
        border="1"
        cellPadding="10"
        style={{
          borderCollapse: "collapse",
          width: "100%",
          marginTop: "10px",
          textAlign: "left"
        }}
      >
        <thead style={{ backgroundColor: "#f2f2f2" }}>
          <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Email</th>
            <th>Company</th>
          </tr>
        </thead>
        <tbody>

```

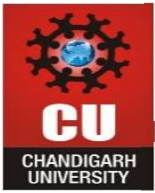
```
        {users.map((user) => (  
            <tr key={user.id}>  
                <td>{user.id}</td>  
                <td>{user.name}</td>  
                <td>{user.email}</td>  
                <td>{user.company.name}</td>  
            </tr>  
        ))}  
    </tbody>  
</table>  
</div>  
);  
}  
  
export default Users;
```

- **ProtectedRoute.jsx**

```
import { Navigate } from "react-router-dom";  
  
function ProtectedRoute({ isAuthenticated, children }) {  
    if (!isAuthenticated) {  
        return <Navigate to="/" replace />;  
    }  
    return children;  
}  
  
export default ProtectedRoute;
```

- **Login.jsx**

```
import { useState } from "react";  
import { useNavigate } from "react-router-dom";  
  
function Login({ setIsAuthenticated }) {  
    const [username, setUsername] = useState("");  
    const [password, setPassword] = useState("");  
    const [error, setError] = useState("");  
    const navigate = useNavigate();  
  
    function handleLogin(e) {  
        e.preventDefault();  
  
        if (username === "admin" && password === "7410") {  
            setIsAuthenticated(true);  
            navigate("/dashboard");  
        } else {
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

```
        setError("Invalid username or password");
    }
}

return (
  <div style={{ padding: "40px", textAlign: "center" }}>
    <h1 style={{ marginBottom: "20px" }}>Admin Login</h1>
    <form onSubmit={handleLogin} style={{ display: "inline-block", textAlign: "left" }}>
      <div style={{ marginBottom: "15px" }}>
        <label style={{ display: "block", marginBottom: "5px" }}>Username:</label>
        <input
          type="text"
          value={username}
          onChange={(e) => setUsername(e.target.value)}
          style={{ padding: "8px", width: "200px" }}
          required
        />
      </div>
      <div style={{ marginBottom: "15px" }}>
        <label style={{ display: "block", marginBottom: "5px" }}>Password:</label>
        <input
          type="password"
          value={password}
          onChange={(e) => setPassword(e.target.value)}
          style={{ padding: "8px", width: "200px" }}
          required
        />
      </div>
      <button
        type="submit"
        style={{
          padding: "10px 20px",
          fontSize: "16px",
          cursor: "pointer",
          backgroundColor: "#4CAF50",
          color: "white",
          border: "none",
          borderRadius: "5px"
        }}
      >
        Login
      </button>
    </form>
    {error && <p style={{ color: "red", marginTop: "15px" }}>{error}</p>}
  </div>
);
}
```

export default Login;

- **NotFound.jsx**

```
function NotFound() {
  return (
    <div style={{ padding: "40px", textAlign: "center" }}>
      <h1 style={{ color: "red" }}>404 - Page Not Found</h1>
    </div>
  );
}

export default NotFound;
```

- **Reports.jsx**

```
import { useEffect, useState } from "react";

function Reports() {
  const [students, setStudents] = useState([]);

  useEffect(() => {
    async function fetchStudents() {
      try {
        const res = await fetch("https://jsonplaceholder.typicode.com/users");
        const data = await res.json();
        setStudents(data);
      } catch (error) {
        console.error("Error fetching students:", error);
      }
    }
    fetchStudents();
  }, []);

  // Generate some mock stats based on student count
  const total = students.length;
  const active = Math.floor(total * 0.7); // ~70% active
  const inactive = total - active;
  const reportsGenerated = total * 2; // pretend 2 reports per student

  return (
    <div style={{ padding: "20px" }}>
      <h2 style={{ marginBottom: "15px" }}>Reports Section</h2>
      <p style={{ marginBottom: "15px" }}>Performance reports displayed here.</p>

      <ul style={{ lineHeight: "1.8", paddingLeft: "20px" }}>
        <li>Total Students: {total}</li>
        <li>Active Students: {active}</li>
        <li>Inactive Students: {inactive}</li>
        <li>Reports Generated: {reportsGenerated}</li>
      </ul>
    </div>
  );
}
```

```
}
```

```
export default Reports;
```

- **App.jsx**

```
import { useState } from "react";
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Login from "../pages/Login";
import Dashboard from "../pages/Dashboard";
import Students from "../pages/Students";
import Reports from "../pages/Reports";
import Users from "../pages/Users";
import NotFound from "../pages/NotFound";
import ProtectedRoute from "../components/ProtectedRoute";

function App() {
  const [isAuthenticated, setIsAuthenticated] = useState(false);

  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Login setIsAuthenticated={setIsAuthenticated} />} />

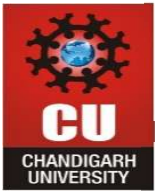
        <Route
          path="/dashboard/*"
          element={
            <ProtectedRoute isAuthenticated={isAuthenticated}>
              <Dashboard setIsAuthenticated={setIsAuthenticated} />
            </ProtectedRoute>
          }
        />
        <Route path="students" element={<Students />} />
        <Route path="reports" element={<Reports />} />
        <Route path="users" element={<Users />} />
      </Routes>
      <Route path="*" element={<NotFound />} />
    </BrowserRouter>
  );
}

export default App;
```

- **Student.jsx**

```
import { useEffect, useState } from "react";

function Students() {
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

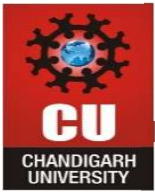
```
const [students, setStudents] = useState([]);

useEffect(() => {
  async function fetchStudents() {
    try {
      const res = await fetch("https://jsonplaceholder.typicode.com/users");
      const data = await res.json();
      setStudents(data);
    } catch (error) {
      console.error("Error fetching students:", error);
    }
  }
  fetchStudents();
}, []);

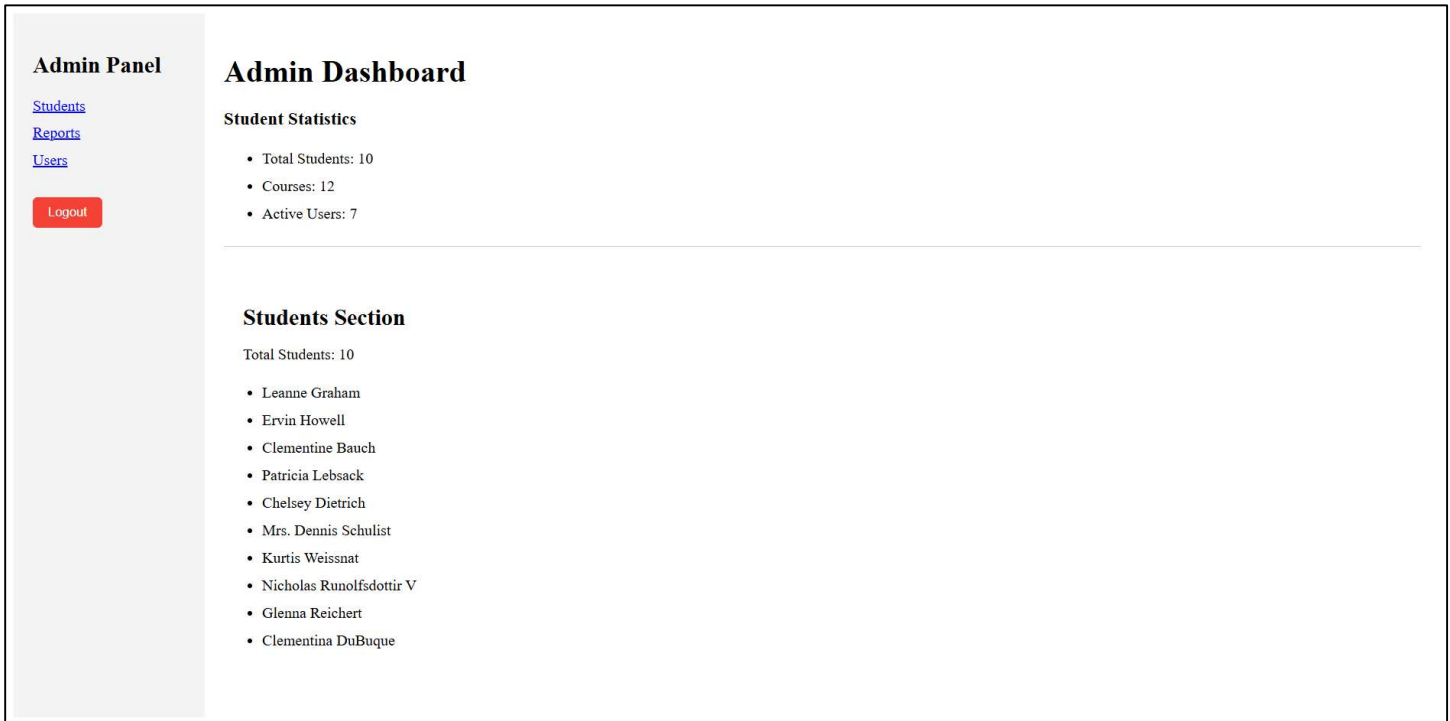
return (
  <div style={{ padding: "20px" }}>
    <h2 style={{ marginBottom: "15px" }}>Students Section</h2>
    <p style={{ marginBottom: "15px" }}>
      Total Students: {students.length}
    </p>

    <ul style={{ lineHeight: "1.8", paddingLeft: "20px" }}>
      {students.map((student) => (
        <li key={student.id}>{student.name}</li>
      ))}
    </ul>
  </div>
);
}
```

export default Students;



4. Output:



5. Learning Outcome:

- Understanding of advanced React Router features including protected and nested routes.
- Ability to implement authentication-based route guards using a custom ProtectedRoute component.
- Skill in using Outlet for nested route rendering and building dashboard layouts.
- Confidence in handling invalid URLs with a custom 404 page.
- Experience in structuring scalable multi-page admin SPAs with authentication.